

S.3 Table: Functional traits for each wasp species based on measurements, laboratory observations and literature source.

Wasp species	Left wing length (mm)	ITS (mm)	Left tibia posterior (mm)	Left mandibula (mm)	Larval diet (own observation)	Nesting material (own observation)
<i>Priochilus</i> sp1	13.83	2.09	5.6	1.55	spider	mud
<i>Priochilus</i> sp2	9.77	1.58	4	1.2	spider	mud
<i>Auplopus</i> sp1	8.7	1.63	24.44	1.2	spider	mud
<i>Auplopus</i> sp2	8.41	1.58	3.5	1.16	spider	mud
<i>Isodontia</i> sp1	11.25	2.64	2.63	1.45	grasshopper	plant_parts
<i>M. angulosa</i>	15.2	3.96	3.68	2.55	larvae	plant_parts
<i>P. alessandroi</i>	11.63	2.43	2.38	1.39	cockroach	plant_parts
<i>P. argentipilis</i>	11.31	2.48	2.77	1.47	larvae	mud
<i>P. brevithorax</i>	11.58	2.55	2.65	1.57	larvae	mud
<i>P. gianelli</i>	11.28	2.98	3.02	1.42	larvae	mud
<i>P. grandis</i>	8.95	2.03	2.04	1.34	larvae	mud
<i>P. guadulpensis</i>	8.11	2.53	2.18	1.37	larvae	mud
<i>P. surinama</i>	9.75	2.24	2.27	1.21	larvae	mud
<i>P. egregium</i>	13.68	2.4	4.44	1.76	cockroach	mud
<i>P. goryanum</i>	13.47	2.7	4.32	1.59	cockroach	mud
<i>P. agile</i>	10.71	2.17	3.6	2.33	cockroach	mud
<i>P. foxii</i>	10.99	2.29	2.32	1.5	cockroach	mud
<i>P. rufipes</i>	13.71	2.86	4.66	3.04	cockroach	mud
<i>T.lactitarse</i>	10.47	2.59	2.86	0.86	spider	mud
<i>T. nitidum</i>	6.62	1.83	1.97	0.75	spider	mud
<i>T. punctivertex</i>	9.79	1.56	1.79	0.69	spider	mud
<i>T. xanthandrum</i>	9.49	1.84	2.61	0.66	spider	mud
<i>Z. proximus</i>	10.91	2.3	2.37	1.32	cockroach	plant_parts
<i>Z. smithii</i>	14.83	3.12	3.31	2.08	cockroach	plant_parts
<i>P. nasidens</i>	8.39	2.66	2.29	1.62	larvae	mud